Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Course: B.Tech.
Code: 100103
Subject: Chemistry
Time: 03 Hours
Full Marks: 70

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Why are Br^+ and CCl_2 electrophiles?

(b) What do you mean by single electrode potential?

(c) Arrange molecular species N_2, N_2^+ and N_2^- in increasing order of stability.

(d) A water sample contains 204 mg of $CaSO_4$ per litre. Calculate the hardness in terms of CaCO3 equivalent?

(e) What is the value of effective nuclear charge (z^*) for last electron of Cu +? (Atomic no. of Cu=29)

(f) Out of Cr^{2+} and Cr^{3+}, which one is stable in aqueous solution?

(g) What is the direction of a reaction, when $\Delta G = 0$

(h) How is the bond order related with dissociation energy?

(i) The specific conductance of a decinormal solution of KCl at 18 °C is 0.0112 $ohm^{-1}cm^{-1}$. The resistance of the cell containing the solution at 18 °C was found to be 55 ohms. What is the cell constant?

(j) A gas expands against vacuum. What is the work done on it?

Q.2 (a) Discuss Heisenberg's uncertainty principle and deduce a mathematical expression involving position and velocity of uncertainty of an electron. **[7]**

(b) Explain dual nature of light and give one example (property/experiment) in favour of its particle nature and wave nature. **[7]**

Q.3 (a) Draw the MO energy-level diagram for O_2 and based on the diagram explain the paramagnetic property observed in O_2. **[7]**

(b) Explain ionization isomerism and linkage isomerism for transition metal complex with an example of each. **[7]**

Q.4 (a) 1 mole of an ideal gas expands isothermally and reversibly from 5 dm^3 to 10 dm^3 at 300 K. Calculate q, w, $\Delta U, \Delta H, \Delta G$ and ΔA. **[7]**

(b) Design electrochemical cells (in proper notation) in which each of the following reaction occurs: **[7]**

$Ce^{4+}(aq) + Fe^{2+}(aq) \rightarrow Ce^{3+}(aq) + Fe^{3+}(aq)$

$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

$HgO(s) + H_2(g) \rightarrow Hg(l) + H_2O(l)$

$Fe^{2+}(aq) + Ag^+(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$

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Q.5 (a) Write the principle for lime soda process for softening of hard water? [7]

(b) A water sample has the following constituents per litre. $CaCO_3$=81 mg, Mg $(HCO_3)_2$ = 75 mg, $CaSo_4$=136 mg, $Mgso_4$=120mg Nacl = 4.7 mg. [7]
Calculate the quantity of temporary and permanent hardness in the water sample.
Calculate the quantity of lime (78% purity) and soda (92% purity) required for softening of 1.5 million litres of the water sample.

Q.6 (a) The internuclear distance of Nacl is 2.36×10^{-10}m. Calculate the reduced mass and moment of inertia of Nacl. (Atomic mass of cl = 35×10^{-3}kg mol^{-1}and Na= 23×10^{-3} kg mol^{-1}) [7]

(b) Write the product for following reactions together with reaction mechanism: [7]
(i) $2CH_3COCH_3 + OH^- \rightarrow$
(ii) CH_3CH_2OH (heated with H_2SO_4) $\rightarrow$

Q.7 (a) Discuss photoelectric effect and explain equations involved with this effect. [7]

(b) How many photons of light having a wavelength of 4000 Å are necessary to provide 1 J of energy? ($h=6.626\times10^{-34}$ J-s) [7]

Q.8 Explain the following: [3½ x4 =14]

(a) S_N1 leads to racemic mixture, whereas S_N2 gives rise to inverted product.
(b) o-nitrophenol easily separates from p-nitrophenol by steam distillation.
(c) The amino group in aniline is o- and p-directing, but amide group is m-directing.
(d) Benzoic acid is weaker than formic acid.

Q.9 (a) Write notes on the following: [3½ x 2 =7]
(i) Optical isomerism of lactic acid
(ii) Diels – Alder reaction.

(b) Differentiate between the following: [3½ x 2 =7]
(i) Enantiomers and diastereomers
(ii) Racemic mixture and meso-compounds

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